MSG No. 134.474 $P4_2'/nn'm$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4,5,6,9,10,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,7,8,11,12,13,14]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[3,4,5,6,9,10,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4,9,10]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,6,15,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[7,8,13,14]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,2,3,4]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[13,14,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,3,5,7]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,4,6,8]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[9,12,13,16]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[10,11,14,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `..2/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3,5,7]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,4,6,8]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[9,12,13,16]`

continued ...

Table 6

No.	position	mapping
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,14,15]`

Table 7: Wyckoff site: `4g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[9,10,15,16]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11,12,13,14]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[3,4]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[5,6]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[7,8]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[13,14]`
8	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,11]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,12]`
3	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[3,9]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[4,10]`
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[8,14]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,12]`
3	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[4,10]`
5	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,15]`

continued ...

Table 10

No.	position	mapping
6	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	[6,16]
7	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[7,13]
8	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[8,14]

Table 11: Wyckoff site: 8k, site symmetry: ..2

No.	position	mapping
1	$[x,\ x,\ 0]$	[1,3]
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	[2,4]
3	$[-x,\ -x,\ 0]$	[5,7]
4	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[6,8]
5	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	[9,12]
6	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[10,11]
7	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[13,16]
8	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[14,15]

Table 12: Wyckoff site: 8l, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	[1,3]
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[2,4]
3	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[5,7]
4	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[6,8]
5	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	[9,12]
6	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	[10,11]
7	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	[13,16]
8	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	[14,15]

Table 13: Wyckoff site: 8m, site symmetry: ..m

No.	position	mapping
1	$[x,\ -x,\ z]$	[1,7]
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[2,8]
3	$[-x,\ x,\ -z]$	[3,5]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[4,6]
5	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[9,16]
6	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[10,15]
7	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[11,14]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[12,13]

Table 14: Wyckoff site: `16n`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[2]`
3	$[y,\, x,\, -z]$	`[3]`
4	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	`[4]`
5	$[-x,\, -y,\, -z]$	`[5]`
6	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	`[6]`
7	$[-y,\, -x,\, z]$	`[7]`
8	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	`[8]`
9	$\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[10]`
11	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	`[12]`
13	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[14]`
15	$\left[-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[15]`
16	$\left[x+\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	`[16]`